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Task Description:

Our long-term goal is a system that uses measures of speech to detect cognitive deficits, personality changes and emotional disturbances during prolonged space flight, allowing crews to invoke countermeasures before performance is severely impaired. The system will monitor conversation, deriving acoustic parameters via automated procedures, without requiring crewmembers to wear intrusive devices. It will protect privacy and mission security because it does not depend on the content of speech, only its acoustics.
Deep-space missions will expose crews to cosmic rays, which may damage brain structures including the basal ganglia and hippocampus. These subcortical structures are linked with cortical regions in circuits regulating motor control, cognition and personality. As models for crews suffering neurological damage we use climbers ascending Mount Everest and patients with Parkinson's disease (PD). At altitudes above Everest Base Camp the low oxygen content can damage the principal basal ganglia output structure, the globus pallidus. The hippocampus also is sensitive to hypoxia. The ascent of Everest also resembles long-term space flight in that a small group in close contact for an extended period must make critical decisions in stressful, life-threatening situations.

PD, which affects basal ganglia function, impairs motor control but also causes deficits in cognition and language comprehension more severe than those in Everest climbers. Thus climbers and PD patients offer complementary Earthbound "space-analogs" for milder and more extreme forms of the dysfunction astronauts may suffer on long-term missions.

The rationale for using speech measures as cognitive indices is that the basal ganglia regulate sequences of speech motor acts as well as of cognitive operations. We can expect the corresponding circuits to show parallel declines in function due to hypoxia, PD or radiological damage. One index we focus on is voice-onset time (VOT). Stop consonants in word-initial position are produced by first closing the lips or placing the tongue against the roof of the mouth. The obstruction then is abruptly released, yielding a "burst" with distinct acoustic properties. Speakers must also adjust the larynx to produce a vowel after the burst. For "voiced" stops like [b] the time between burst and vowel – the VOT – is less than ~30 msec, while for "voiceless" stops like [p] it is greater than ~50 msec. Hence VOT production requires precise control of the sequence of oral and laryngeal actions. PD patients show reduced differences between VOTs for voiced and voiceless stops. Such VOT convergence correlates with deficits in sentence comprehension.

On Everest we obtain measurements of VOT and other speech parameters, e.g., vowel duration, by using radios to record climbers reading words at different altitudes. We have measured cognition by various tests. Climbers carry picture booklets so that we can administer, by radio, sentence-picture matching tests of language comprehension. To measure cognitive flexibility we give the Odd-Man-Out test (OMO) and, in 2004, the Wisconsin Card Sorting Task. These require subjects to select stimuli using one of several possible criteria and then shift criteria following feedback. In 2003 we added the Mini-Cog test battery developed by Stephen Kosslyn's NSBRI group at Harvard. This battery, which tests cognitive capacities including verbal and spatial working memory, is implemented on Palm Pilot PDAs that climbers carry. In 2004 a group of climbers also took an implicit contextual learning test that reflects hippocampal function.

Results over successive expeditions bear out our hypotheses. At high altitudes many climbers show VOT convergence and increased vowel duration. Crucially, these changes track impairment on cognitive tasks that depend on basal ganglia function. Greater VOT separation decreases were seen for larger increases in sentence comprehension response time (RT) with altitude. In 73% of cases, the presence (or absence) of reduced VOT separation signaled the presence (or absence) of deficits in comprehension RT or OMO performance. VOT separation had a similar 71% "hit rate" in discriminating impaired from unimpaired performance on Mini-Cog tests of working memory or vigilance. Vowel duration had a hit rate of 85% for sentence comprehension or OMO and was significantly correlated with verbal working memory RT. Our speech metrics track severe cognitive deficits as well. One climber developed profound speech motor sequencing deficits and similarly extreme deficits in cognitive flexibility. Advised of his condition, he nonetheless proceeded to climb upwards, alone, through a storm; two days later he fell to his death.

Preliminary analyses indicate that hypoxia impaired hippocampal function. Like amnesics, climbers at higher altitudes failed to show implicit contextual learning. Altitude did not affect performance on tasks that do not place demands on basal ganglia or hippocampal circuitry.

Our PD studies support our hypotheses on the basal ganglia's role in understanding language. By tracking eye movements as subjects do a sentence-picture matching task we have shown that patients with high OMO error rates also exhibit impaired processing at points in sentences that require linguistic "set-shifting."

Our techniques may be used to monitor PD and have potential for detecting early memory loss in Alzheimer's disease, enabling interventions that could slow further decline. We are exploring the assessment of verbal apraxia in children, a disorder affecting linguistic and cognitive development that may result from hypoxia during birth. Our techniques have potential applications in general aviation, where hypoxia has led to disasters.

In this project's next phase we will apply speech analysis techniques to study task-induced stress in synergy with Kosslyn's group and David Dinges's NSBRI group at the University of Pennsylvania.

The overarching goal of our research is the development of a monitoring system that predicts cognitive dysfunctions associated with deep space travel conditions. Cosmic radiation associated with deep space travel is likely to affect subcortical structures in the brain in a way similar to oxygen deprivation by compromising the function of basal ganglia circuits involved in both motor behavior and cognition. Speech is affected by small disturbances to motor circuits. By studying climbers on Mount Everest, a stressful, life-threatening environment analogous to the situation astronauts will face in deep space, we aim to develop and validate techniques for monitoring effects of neurological impairment via minute yet measurable changes in speech production. Analyses of data from earlier expeditions have revealed aspects of speech that correlate with specific cognitive impairments. We have refined our battery of cognitive measurements, most recently incorporating improved tests administered via handheld Palm Pilots carried by climbers as they ascend Everest. This technology has made it possible to collect data at higher locations where contact with research assistants has been difficult or impossible. The wider range of functions assessed by these tests has allowed a more detailed and complete synopsis of subtle cognitive changes that occur with hypoxia. The Palm Pilots also serve as additional speech recording tools, providing clearer recordings in severe weather conditions that previously limited radio transmission.

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The techniques we are developing for unobtrusively monitoring cognitive status via automated measurement of speech parameters may have applications in general aviation. Systems based on these techniques could be used to monitor air crews for gradual effects of partial or slow failure of aircraft pressurization systems. The hypoxia resulting from such depressurization – which degrades cognitive function so that crew members not only are unable to perform their tasks but fail to notice their own impairment – has led to flight disasters in the past. Speech-monitoring techniques might also be useful in predicting the onset of Acute Mountain Sickness before it occurs in personnel deployed to high mountain regions. A speech-based system may also be useful in monitoring motor and cognitive dysfunction resulting from stress and sleep deprivation. Our project's techniques have already been used to assess the efficacy of new surgical procedures for the treatment of Parkinson's disease. They can be used to monitor disease state in Parkinson's. They may also provide instruments that can detect memory loss in the early stages of Alzheimer's disease. Such early detection would permit clinicians to take maximal advantage of therapies, now under development, that can delay or even arrest further decline. In addition to Parkinson's disease, our techniques may have application to the diagnosis, assessment, and treatment of other human pathologies stemming from impaired dopaminergic basal ganglia function in neural circuits regulating speech production, cognition, and personality. These include not only neurodegenerative diseases but also the results of acute insult. For example, hypoxia during birth can lead to verbal apraxia in children – a syndrome where speech motor and orofacial motor control is degraded and which can result in cognitive and linguistic deficits.
Task Progress:	Results from earlier Everest expeditions indicated that aspects of speech correlated with specific cognitive impairments. The most important progress we have made over the last funded year was in obtaining further data and performing analyses that lend firmer statistical support for these earlier findings. These results are described under Main Findings. In previous years cognitive assessments were based on two standard tests of sentence comprehension and cognitive set-switching. In 2003 we added the Mini-Cog test battery that Kosslyn and colleagues have been developing in their NSBRI-supported research. Mini-Cog provided nine new tests of functions such as vigilance, divided attention, and spatial and verbal working memory. The tests are programmed onto handheld Palm PDAs carried by the climbers and can be self-administered; the PDAs also serve as speech recorders. This technology allowed testing and recording even when weather obstructed radio transmission to researchers at Base Camp or kept subjects from leaving their tents to get radios. It also enabled data collection at higher locations where contact with researchers had been difficult or impossible. Our field experience suggested a number of modifications to the Mini-Cog battery, which Kosslyn's group has implemented. In 2003 some data was lost when PDAs froze; for 2004 the system was modified to transfer data and recordings to digital cards that preserve the data in this circumstance. In 2004 we expanded the range of brain systems and cognitive functions studied by giving a subgroup of climbers a test of implicit contextual learning. This test reflects the function of memory systems dependent on the hippocampus and has been shown to be sensitive to deficits in amnesics. We also replaced our previous test of set-shifting, the Odd-Man-Out test, with the more sensitive Wisconsin Card Sorting Task. To test sentence comprehension our initial studies used the Rhode Island Test of Language Structure (RITLS), a sentence-picture matching test designed for use with hearing-impaired children. In 2002 we replaced the RITLS with a test encompassing a better-controlled set of linguistic structures. In 2003 and 2004 we narrowed the range of structures tested to focus on those found the previous year to be most sensitive to hypoxia (while retaining some easier types for control purposes). The shortened test retains its sensitivity while being less time-consuming. We have also obtained speech recordings during stressful situations, which we plan, in the next phase of our research, to compare with speech samples reflecting task-induced stress obtained in the NSBRI-supported study of Dinges and colleagues at the University of Pennsylvania. These advances in test design and speech measures have allowed a more detailed and complete synopsis of subtle changes that occur with hypoxia and have moved our voice measures and cognitive tests substantially closer to the stage of being useful in flight testing.
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